Week 9 — Photonic Quantum Computing Platforms and Quantum Networks

EEQT30001 Quantum Technology and Engineering, Unit 2 (Quantum Optics and Quantum Sensing)

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Learning objectives (Week 9)

1. Draw up the photonic qubit's balance sheet — why photons barely decohere and barely interact (Maxwell linearity) — and explain how KLM turns *measurement* into an effective nonlinearity: probabilistic heralded gates, and the cluster-state (MBQC) and fusion-based architectures built on that idea.
2. Explain Gaussian boson sampling as "computing with squeezed vacuum": why sampling the output of a linear interferometer is classically hard (hafnian), recount the Jiuzhang–Borealis–Aurora milestones, and distinguish quantum *advantage* from quantum *usefulness*.
3. Explain why fiber loss plus no-cloning rules out amplifiers and forces the quantum-repeater architecture (segment entanglement + swapping + memories), and place today's demonstrations — Micius, Delft, metro-fiber networks — on that roadmap.

Reading: Kok *et al.*, Rev. Mod. Phys. **79**, 135 (2007), Secs. I–II; Slussarenko & Pryde, Appl. Phys. Rev. **6**, 041303 (2019). GBS context: QO_note_6, Sec. 1. Networks: Azuma *et al.*, Rev. Mod. Phys. **95**, 045006 (2023), Sec. I.

Opening puzzle. The photon is the best qubit anyone has ever found: it does not decohere, it moves at the speed of light, and it works at room temperature — all because it interacts with almost nothing. But wait. Computation *requires* interaction: a two-qubit gate is precisely one qubit changing another. How do you compute with a particle that does not talk to anything? This week gives the field's three answers: steal the interaction from measurement (Sec. 1), pick a problem where passive interference is already enough (Sec. 2), and embrace the non-interaction — if the photon talks to nothing, nothing can stop it from carrying quantum states across a city or up to a satellite (Sec. 3).

1 The DV Route: Measurement Replaces Interaction

1.1 The photonic qubit's balance sheet

Assets	Liability
negligible decoherence (no charge, no rest mass, weak coupling to the environment) flies at c — communication is built in room-temperature qubit carrier compatible with silicon-photonics fabrication	**photons do not interact with each other** ⇒ deterministic two-photon gates are unavailable

Circle the right-hand column. The entire shape of photonic quantum computing — every architecture in this section — is a response to that single cell.

Why photons do not interact. In vacuum and in linear media, Maxwell's equations are linear: superposition is exact, and two crossing laser beams pass through each other unchanged. (Quantum electrodynamics does permit photon–photon scattering via virtual electron–positron pairs, but at optical energies the rate is so small it has never been observed with light alone.) The engineering escape would be a $\chi^{(3)}$ (Kerr) medium, where one beam shifts the phase of another — but at the *single-photon* level the available nonlinear phase in conventional materials falls short of the π a gate needs by many orders of magnitude [2]. Nature does not provide a photon–photon gate. The field had to invent one.

1.2 Encoding one photon into one qubit

Three standard encodings of a dual-mode qubit (Fig. 1): polarization ($|H\rangle, |V\rangle$), *dual rail* (which path or waveguide the photon occupies), and *time bin* (early or late slot). All three are two-mode encodings, interconverted by polarizing beam splitters and delay lines, and everything from Week 8 applies verbatim: single-qubit gates are wave plates, beam splitters, and phase shifters — linear optics does arbitrary single-qubit rotations easily. Only the *two*-qubit gate is missing.

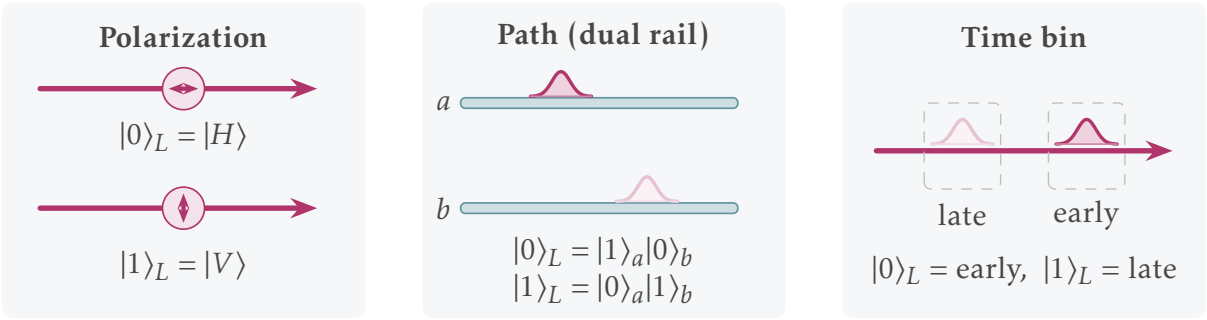


Figure 1: Three ways to write one qubit on one photon. In each case the logical information lives in *which mode* the photon occupies.

1.3 KLM: the measurement-induced nonlinearity

In 2001 Knill, Laflamme, and Milburn showed that linear optics *plus* ancilla photons *plus* photon counting is enough for universal quantum computation [1]. The trick: you cannot make two photons talk, but a measurement can. Mix the signal modes with ancilla photons on a network of beam splitters, then detect the ancilla modes (Fig. 2). Conditioned on a particular detection pattern, the surviving signal state has undergone a transformation that no linear network could apply deterministically. The elementary example is the *nonlinear sign* (NS) gate,

$$\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle \longrightarrow \alpha|0\rangle + \beta|1\rangle - \gamma|2\rangle, \tag{1.1}$$

implemented with two ancilla modes — one carrying a single photon, one in vacuum, both detected at the output — and success probability 1/4; two NS gates inside a Hong–Ou–Mandel-style interferometer give a controlled-Z gate with success probability 1/16 [1, 2]. The gate is probabilistic but *heralded*: a specific ancilla click pattern lights up exactly when the gate worked, so you always know.

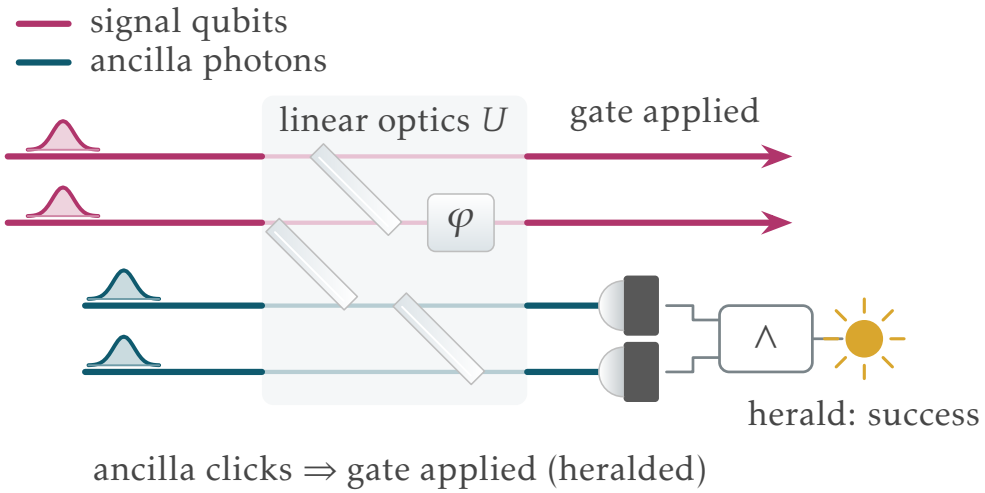


Figure 2: Measurement-induced nonlinearity. Signal qubits and ancilla photons mix in a linear network; a coincidence of ancilla detections heralds success, and only then is the (nonlinear) gate accepted. Layout after Kok *et al.*, Rev. Mod. Phys. **79**, 135 (2007).

The cost, and the second trick. Cascading 1/16 gates naively is hopeless — success probabilities multiply, so the yield of a deep circuit dies exponentially. KLM's rescue is Week 8's teleportation, promoted from stunt to compiler tool: apply the probabilistic gate *offline* to one half of an entangled resource pair, retry until it succeeds, and only then teleport the computational qubit through the pre-gated pair. Failures burn resource states, never data, and larger ancilla states push the teleportation success toward one [1]. We only state the idea — the resource bookkeeping is beyond this course — but the slogan deserves its first appearance here: *measurement is not a passive readout; it is an active computational resource.*

1.4 Cluster states: computing by measuring

Measurement-based quantum computing (MBQC) takes the slogan to its logical conclusion [3]. First entangle a large register of qubits into a standard *cluster state* (a lattice with controlled-Z bonds between neighbors); then compute using *only single-qubit measurements*, column by column (Fig. 3, left). The choice of measurement angles *is* the program; earlier outcomes steer later measurement bases. No two-qubit gate is ever applied during the computation — all the entanglement was paid for up front, in the preparation of the cluster. For photonics this is the perfect division of labor: the hard entangling step moves into *state preparation*, where probabilistic-but-heralded tricks are acceptable (retry until success), while the computation itself needs only the single-photon measurements photons are best at.

1.5 Fusion-based quantum computing: the PsiQuantum bet

Growing one giant cluster photon-by-photon is fragile. The fusion-based architecture [4] instead mass-produces small, identical *seed states* (few-photon entangled states, the only things a chip must generate) and stitches them together with *fusion* measurements — two-photon Bell-type measurements that consume the measured pair and leave an entangling bond between the seeds (Fig. 3, right). A failed fusion is a *heralded erasure* at a known location, exactly the error a fault-tolerant code handles best, so error correction is native to the architecture rather than bolted

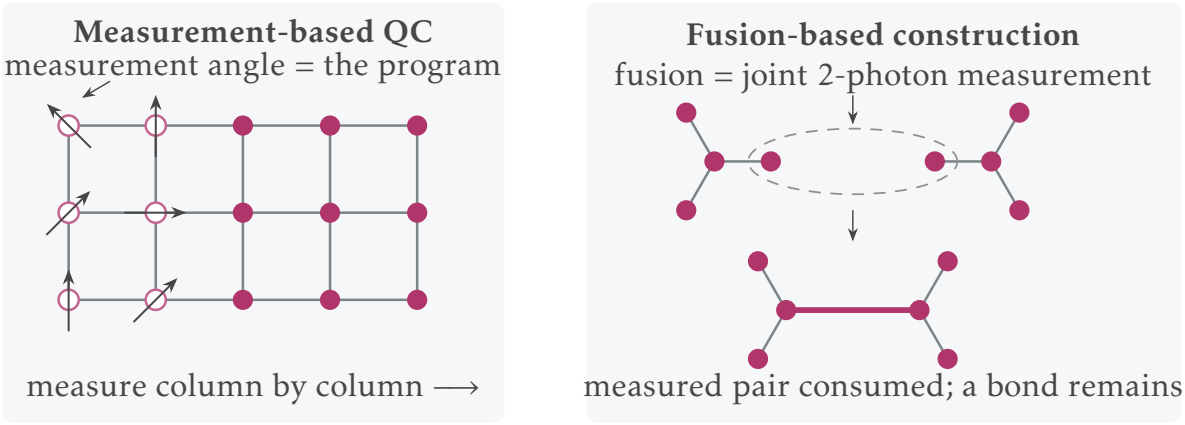


Figure 3: Left: MBQC — entangle first (bonds), then compute by measuring single qubits at program-defined angles. Right: fusion-based construction — small seed states are stitched into the lattice by two-photon fusion measurements; the measured pair is consumed and leaves a bond (after Bartolucci *et al.*, Nat. Commun. **14**, 912 (2023)).

on.

Industry anchor. PsiQuantum is the purest expression of this route: it builds no NISQ-scale machine; its hardware so far is fab-made silicon (the Omega chipset [11]), and it aims directly at a fault-tolerant, million-qubit machine, with chips manufactured in a commercial semiconductor fab (partnership with GlobalFoundries) and utility-scale sites announced near Brisbane (A$940 million, ≈US$0.6 billion, in Australian federal and Queensland government support, 2024; the site has since moved to Moreton Bay) and in Chicago.[1] Why dare to skip the NISQ stage? Because the wager is not on physics milestones but on *manufacturability*: if qubits can be printed by the same process line that already ships datacenter optical interconnects, then scaling is a fab problem, not a laboratory problem. The contrast with superconducting platforms is stark — no dilution refrigerators; the photons themselves live at room temperature, and only the single-photon detectors (Week 7's SNSPDs) need compact few-kelvin cryostats.

In-class discussion 1. *"Photons barely interact with each other" is the liability of this whole section. Is it the same fact as "optical fiber signals do not interfere with each other", which telecom engineers list as an asset?* Yes — both are the linearity of Maxwell's equations in the medium. One community curses it (no two-photon gate), the other banks on it (wavelength-division multiplexing works). Nature gave no nonlinearity at the single-photon scale; KLM's answer is to steal one from measurement.

Summary: Section 1

Photons make superb single qubits (no decoherence, room temperature, fab-compatible) but refuse to interact, so deterministic two-photon gates do not exist. KLM 2001: linear optics + ancilla photons + photon counting yields heralded probabilistic gates, Eq. (1.1), and teleportation moves the failures offline. MBQC recasts computation as single-qubit measurements on a pre-entangled cluster state; fusion-based architectures build that cluster from mass-produced seed states joined by heralded fusion measurements, with fault tolerance built into the construction. Measurement is the resource throughout.

[1] Industry figures in this note are as of 2026-09 and are re-verified before class; sources are cited in the LaTeX comments beside each claim.

2 The CV Route: Computing with Squeezed Vacuum

2.1 Switching languages

Section 1 counted photons. The continuous-variable (CV) route measures fields: the information carriers are the quadratures $X = (a + a^\dagger)/\sqrt{2}$ and $P = (a - a^\dagger)/(i\sqrt{2})$ of Week 7, the workhorse measurement is homodyne detection, and the quantum resource is the squeezed vacuum $S(r)|0\rangle$ — deterministically produced by a pumped $\chi^{(2)}$ crystal (Week 8's parametric amplifier). No heralding, no waiting: every pulse delivers a squeezed state. That determinism is the CV route's opening move, and it leads to this week's installment of our running dark line: *the input to the quantum-advantage experiments below is squeezed vacuum — they are, literally, computing with vacuum.*

2.2 Gaussian boson sampling in a nutshell

Boson sampling is a quantum Galton board (Fig. 4). In the classical board, balls rattle through pegs and pile up in a binomial distribution — easy to predict. Replace the balls by *indistinguishable* photons and the pegs by a mesh of beam splitters and phase shifters implementing a Haar-random unitary U on M modes, place photon counters at the outputs, and the probability of a click pattern $\mathbf{n} = (n_1, \ldots, n_M)$ is no longer binomial: for single-photon inputs it is proportional to $|\mathrm{Per}(U_S)|^2$, the permanent of an $N \times N$ submatrix of U selected by the pattern. Computing permanents is #P-hard, and the interference of all $N!$ photon paths cannot be factorized, so for ideal (lossless) boson sampling with $N \gtrsim 50$ photons no classical computer is believed able to sample from this distribution [6]. *Gaussian* boson sampling (GBS) makes one substitution: the inputs are squeezed-vacuum states instead of heralded single photons, and the permanent becomes the *hafnian* — a cousin of the permanent, just as hard — with

$$\Pr(\mathbf{n}) \propto \left|\mathrm{Haf}(A_S)\right|^2, \tag{2.1}$$

where A_S is the submatrix, fixed by the click pattern, of a matrix A built from U and the squeezing parameters. Because squeezed vacuum is deterministic while heralded single photons are not, GBS is the version that scales, and it underlies every photonic quantum-advantage demonstration to date. Boson sampling is *not* a universal computer — it is a minimal, purely passive experiment designed to expose one thing: the computational power of bosonic interference.

2.3 Milestones: Jiuzhang, Borealis, Aurora

- **Jiuzhang** (USTC, 2020): 76-photon GBS through a 100-mode interferometer — the first claim of photonic quantum advantage [7]. **Jiuzhang 3.0** (2023) upgraded to pseudo-photon-number-resolving detection and registered up to 255 clicks [8].
- **Borealis** (Xanadu, 2022): 216 squeezed *time-bin* modes in a dynamically programmable loop interferometer — quantum advantage on a machine users could program over the cloud [9].
- **Aurora** (Xanadu, 2025): 35 photonic chips linked by 13 km of fiber across four server racks — not a bigger sampler but a *modular, networked* architecture aimed at fault tolerance [10].

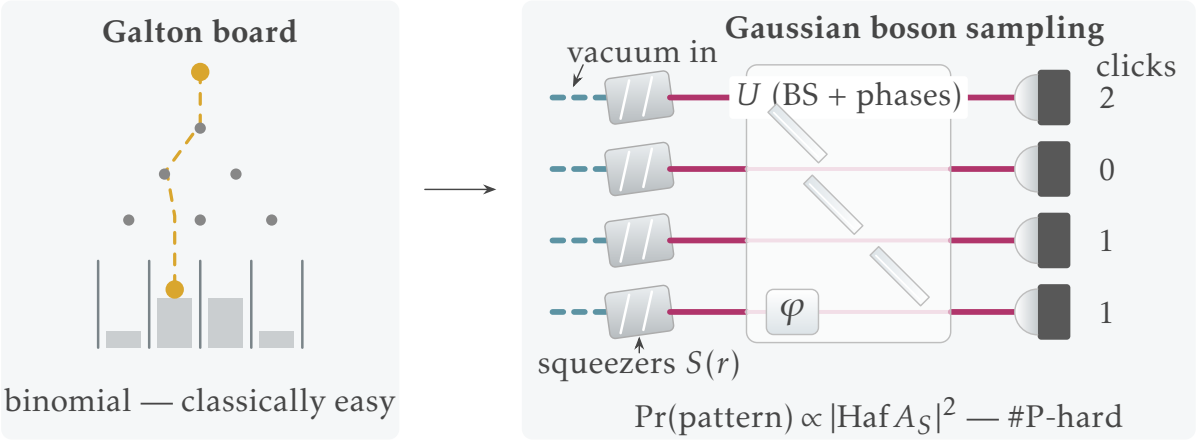


Figure 4: Balls → photons, pegs → beam splitters. Left: classical Galton board, binomial output. Right: GBS — squeezed vacuum enters a programmable interferometer U; photon-number-resolved click patterns are distributed according to a #P-hard hafnian, Eq. (2.1). Layout after Madsen *et al.*, Nature **606**, 75 (2022), Fig. 1.

2.4 Industry snapshot 2026

Xanadu's road from Borealis runs through error correction: after the Aurora modular system (January 2025), a June 2025 milestone demonstrated on-chip generation of GKP states [12] (the first optical GKP states had been made in free space a year earlier [13]) — in one sentence, GKP [14] is the encoding that folds a continuous-variable mode into a discrete qubit, and one sentence is all the depth we need. Then, in March 2026, a SPAC merger (with Crane Harbor Acquisition Corp.) that listed Xanadu on Nasdaq and the TSX under the ticker XNDU, raising about US$302 million — the first pure-play photonic quantum-computing company to trade publicly.

	PsiQuantum	**Xanadu**	**Quandela / ORCA**	**China**
encoding	DV (dual rail)	CV (GBS → GKP)	DV (Quandela: QD; ORCA: SPDC + fibre loops)	GBS (Jiuzhang); CIM-type (QBoson)
architecture	fusion-based lattice	modular GBS network	processors around single-photon sources	USTC/Jiuzhang line (GBS); QBoson special-purpose optical machines
signature bet	fab-first, no NISQ machine (GlobalFoundries)	error-corrected CV (Aurora, XNDU)	source quality (Quandela); room-temperature fibre racks (ORCA)	advantage demos (USTC); Shenzhen plant (QBoson)

2.5 The honest page: what is GBS actually for?

Quantum *advantage* is a complexity statement: "no classical computer can reproduce these samples in reasonable time." It is not a statement of *usefulness*, and the two keep being conflated. Proposed applications — molecular vibronic spectra, graph problems, GBS-based machine-

learning feature maps — remain contested territory rather than demonstrated wins. Meanwhile the classical side keeps counterattacking: tensor-network simulators that exploit realistic photon loss have reproduced the statistics of the largest published GBS experiments at modest cost [15], and the cross-entropy benchmarks used to certify advantage can themselves be spoofed [16]. The advantage frontier is real but it *moves*. This page is the most honest one in the course: when a field is hot, the scientific skill you need most is to keep the claims and the evidence in separate, clearly labeled boxes.

In-class discussion 2. *What exactly is the difference between "quantum advantage" and "quantum usefulness"?* Advantage: a sampling task, chosen for hardness, done faster than the best known classical simulation — a complexity milestone. Usefulness: beating classical methods on a problem somebody outside quantum computing already wants solved. Jiuzhang and Borealis achieved the first; nobody has yet demonstrated the second, and saying so plainly is part of doing science.

Summary: Section 2

The CV route computes with the quadratures of Week 7; its resource, squeezed vacuum, is produced deterministically — the dark line of the unit becomes literal, since GBS machines take vacuum as input. Sampling the output of a random interferometer is #P-hard (permanent for single photons, hafnian for squeezed inputs, Eq. (2.1)); Jiuzhang and Borealis turned that hardness into quantum-advantage demonstrations, and Aurora turned the platform into a modular, error-correction-oriented roadmap (GKP). Advantage, however, is a complexity claim — classical simulators exploiting loss keep clawing experiments back, and usefulness remains an open question.

3 Quantum Networks: When Not Interacting Becomes the Point

3.1 The shape of the problem

Flip Section 1's liability: a particle that talks to nothing is the perfect courier. One obstacle remains — loss. Telecom fiber attenuates by $\alpha \approx 0.2\,\mathrm{dB/km}$ at 1550 nm, so direct transmission succeeds with probability

$$P(L) = 10^{-\alpha L/10}: \qquad P(500\,\mathrm{km}) \approx 10^{-10}, \qquad P(1000\,\mathrm{km}) \approx 10^{-20}. \tag{3.1}$$

At 10^{-10}, a 10-GHz single-photon source delivers about one photon per second; at 10^{-20}, one photon every ~300 years (Fig. 5) [17]. Classical telecom solves this with amplifiers every ~80 km — measure the dying signal, copy it louder. But Week 8 proved the no-cloning theorem: an unknown quantum state cannot be copied, and a deterministic amplifier would be exactly a copier. So the problem has a peculiar shape: *the signal dies with distance, and you are forbidden to copy it.*

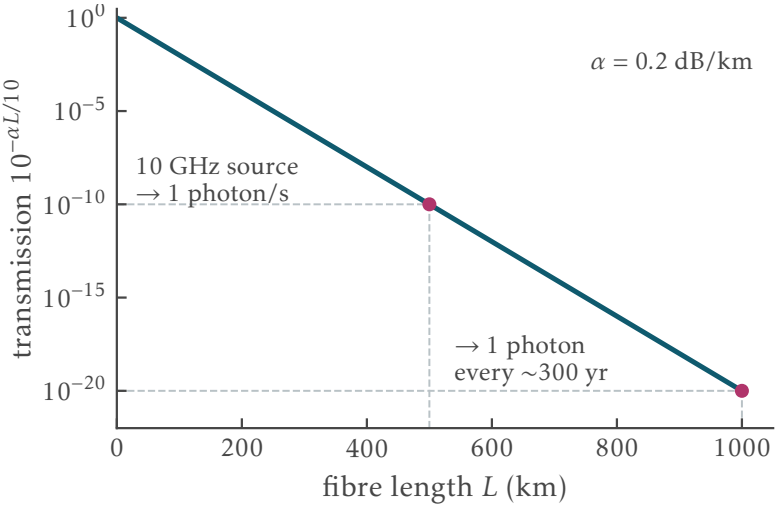


Figure 5: Direct transmission through telecom fiber, Eq. (3.1). The exponential wall is why long-distance quantum communication needs repeaters or satellites, not better lasers.

3.2 Quantum repeaters: entanglement relays

The escape [17] uses the two tools Week 8 handed us (Fig. 6). Cut the distance L into short segments. (1) Generate entanglement across each segment; this is lossy and probabilistic, but each attempt is *heralded*, and a success is parked in a *quantum memory* at each node while neighboring segments catch up. (2) Perform a Bell-state measurement on the two memories at a middle node — *entanglement swapping*, Zeilinger's technique — which splices two short entangled links into one long one. It is Week 8's teleportation applied to half of the left-hand pair: the BSM plus two classical bits hand that entanglement on, so two memories that never interacted end up entangled. Repeat up the line until Alice and Bob share one end-to-end entangled pair, then use it: teleport a qubit or run entanglement-based QKD. Note what never happens: no quantum signal ever crosses the full distance, and nothing is amplified. Entanglement, classical communication, and memories replace the forbidden copier, and the resource cost scales polynomially with L instead of exponentially.

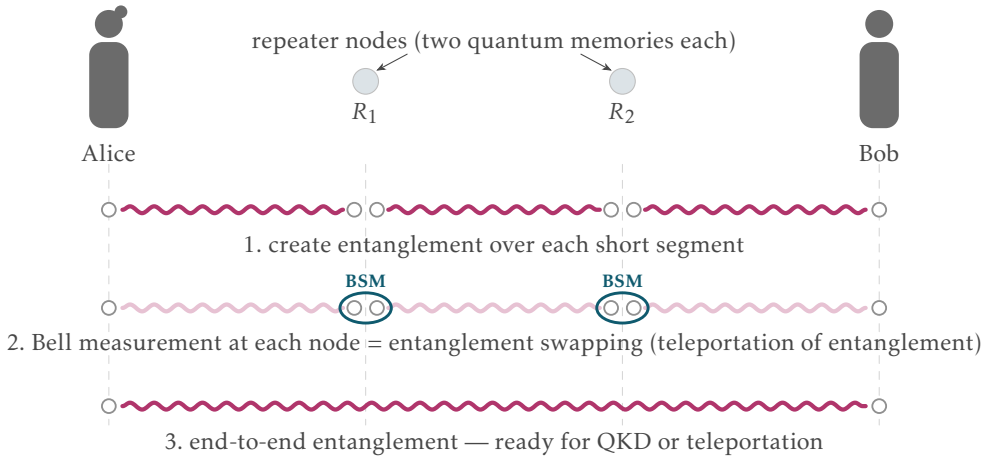


Figure 6: A quantum repeater is an entanglement relay: entangle each short segment (heralded, memories hold the successes), swap at the nodes, and only end-to-end entanglement — never the signal itself — spans the distance. Architecture after Azuma *et al.*, Rev. Mod. Phys. **95**, 045006 (2023).

3.3 Textbook-grade milestones

Micius (2017): the Chinese satellite distributed entangled photon pairs to ground stations 1200 km apart [18] — beating the fiber wall by leaving the fiber; free-space loss grows only quadratically (diffraction) rather than exponentially, and most of the path is vacuum. **Delft** (2021–22): a three-node network of nitrogen-vacancy processors demonstrated genuinely networked protocols — multi-node entanglement distribution [20] and then teleportation between *non-adjacent* nodes [21], i.e., between two parties with no direct link, exactly the repeater primitive of Fig. 6 run on real hardware.

3.4 In progress right now (2025–2026)

None of this is future tense anymore. In New York City, a Qunnect–NYU–Cisco collaboration ran three-node entanglement swapping over 17.6 km of *live commercial* telecom fiber at about 1.5 swaps per second (2026); Photonic Inc. and the carrier TELUS teleported qubit states across 30 km of deployed fiber in a metropolitan network (2026). At the infrastructure scale: China's 2000-km Beijing–Shanghai backbone carries QKD traffic today [19] (with *trusted relay* nodes — honest caveat: that is classical trust at each relay, not yet entanglement swapping); the EU is building EuroQCI; the US DOE runs a national quantum-network program. And the market is consolidating — IonQ bought a controlling stake in the QKD pioneer ID Quantique (May 2025) and acquired the network-hardware maker Qubitekk (January 2025). Quantum networking has left the optical table and moved into telecom conduits.

3.5 Bridge to Week 10: the other killer application

Secrecy is not the only reason to entangle distant nodes. A network of entangled sensors — clocks, telescopes, gravimeters — can beat the precision of the same sensors operating alone; entangled atomic-clock networks and long-baseline optical telescopes sit in the later stages of the six-stage quantum-internet roadmap of Wehner *et al.* [22] (the quantum-memory and few-qubit fault-tolerant stages). What exactly does entanglement buy a *measurement*? What sets the precision limit in the first place, and who ordered it — Heisenberg or statistics? That is Week 10, where the unit's dark line closes: the vacuum is not empty, it is a resource, it can compute — and next week, it can be squeezed and *designed*.

In-class discussion 3. *Classical repeaters amplify the signal every ∼80 km. Why can't a quantum network do the same?* No-cloning: a deterministic amplifier that restores an unknown qubit would be a copier, which unitarity forbids (Week 8's two-line proof). Hence the entire architecture of Fig. 6 — relay *entanglement* by swapping, never the signal by amplification. The take-away sentence of the week applies here too: the repeater's central act, the Bell-state measurement, is measurement working as a resource — KLM's lesson, stretched across a continent.

Summary: Section 3

Fiber loss is exponential, Eq. (3.1), and no-cloning forbids amplification, so quantum networks relay entanglement instead of signals: heralded segment entanglement, quantum memories, and entanglement swapping at the nodes. Micius took entanglement to 1200 km via satellite; Delft ran the first true three-node network protocols; 2025–26 demonstrations

moved entanglement swapping and teleportation onto live commercial metro fiber, while trusted-relay QKD backbones, EuroQCI, and DOE programs build out infrastructure. The same networks' second application — distributed quantum sensing — is where Week 10 begins.

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